

# From Defect Detection to Flight Clearance: a Bayesian SHM Stack for Reusable Launch Vehicles

Stage 0–3 implemented and verified on CFRP thermoelastic fields

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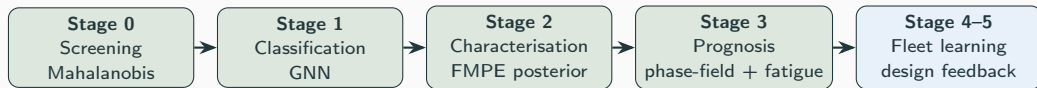
**Expendable:** “will it survive one flight?”

**Reusable:** “is this vehicle OK to fly *again* — and how many flights remain?”

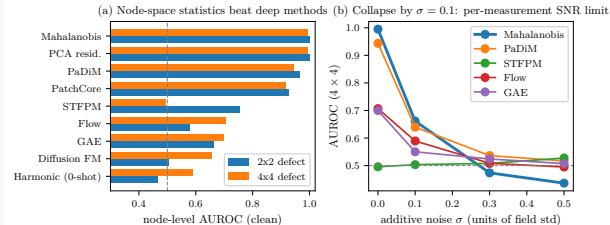
- Same vehicle re-inspected after every flight  
⇒ *fixed geometry, fixed individual* is the norm
- SpaceX (empirically): condition-based monitoring, staged certification 20 → 40 flights, fleet-leader strategy (B1067: 35 flights, 06/2026)

## This work

Formalise that practice as a measurable, Bayesian, physics-based pipeline — applicable from the *first* article.



- **Green = implemented and tested** (2026-06-11, two repos, all committed)
- One data contract end-to-end: posterior  $\theta = (c_x, c_y, \text{layer}, \text{size})$  flows from Stage 2 directly into Stage 3



## Node-level AUROC (clean):

- per-node Mahalanobis: **0.999 / 0.995** (2x2/4x4)
- deep methods: 0.50–0.75

## SNR limit

All methods collapse at noise

$\sigma=0.1$ :

defect amplitude  $< 0.1\sigma$ . Lock-in IR averages  $10^3$  cycles  $\Rightarrow$  requirement, not blocker.

## Fixed geometry = template statistics

Re-inspecting the *same* vehicle: healthy manifold is a point cloud around one template. Per-node  $(\mu_i, \sigma_i)$  is the correct inductive bias.

## Complementary weaknesses

|                     | Mahalanobis | GNN          |
|---------------------|-------------|--------------|
| unknown defect size | open        | collapses    |
| noise (12%)         | dies        | robust       |
| semantic class      | —           | only option  |
| new airframe        | needs refs  | transferable |

⇒ statistics dominate as a vehicle accumulates flight history; learned models bootstrap *new* airframes and repairs.

Flow-matching posterior estimation, trained on the *existing* FEM set (20k pairs, zero new simulations):

| parameter      | RMSE  | $R^2$            | 90%-coverage |
|----------------|-------|------------------|--------------|
| position $c_x$ | 0.055 | 0.78             | <b>0.944</b> |
| position $c_y$ | 0.021 | 0.83             | <b>0.940</b> |
| size (3-class) | —     | 0.73 (acc. 0.82) | 0.894        |
| layer (depth)  | 2.98  | 0.63             | 0.870        |

- Coverage  $\approx$  target 0.90  $\Rightarrow$  **calibrated** posteriors, safe to propagate downstream
- Depth weakly identifiable — a physics limit of surface thermoelastic measurement, reported honestly

## Model (12/12 tests green):

- multi-ply anisotropic AT2,  $A = I + \beta a \otimes a$  per ply
- interface bands ( $G_c^{int} / G_c^{ply} = 0.4$ , DCB-anchored)
- Carrara-2020 fatigue: history  $\bar{\alpha}$  degrades  $G_c$   
 $\Rightarrow$  Wöhler-like growth under repeated sub-critical loads

## Verified physics:

- fiber at  $30^\circ \rightarrow$  crack deflects to  $23.3^\circ$
- low- $G_c$  interface captures crack (73 vs 0 cells)
- $P(\text{growth})$  monotone in load:  
 $0 \rightarrow \frac{2}{3} \rightarrow 1$

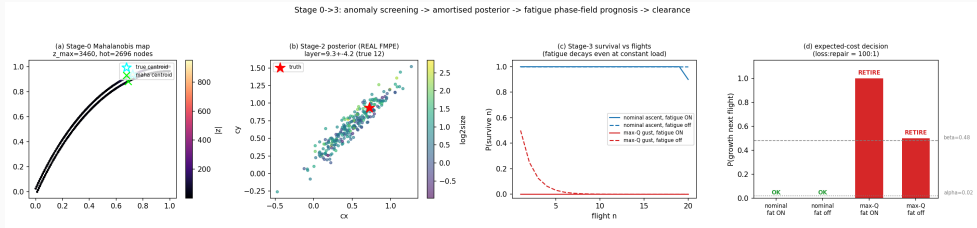


Stage 0→3 run: held-out 4x4 field → Mahalanobis map → *trained* FMPE posterior → fatigue-aware clearance (20-flight horizon).

| load case      | load | fatigue   | $P(\text{grow})$ | $\mathbb{E}[\text{flights}]$ | decision      |
|----------------|------|-----------|------------------|------------------------------|---------------|
| nominal ascent | 0.06 | <b>ON</b> | 0.00             | <b>19.9</b>                  | <b>OK</b>     |
| nominal ascent | 0.06 | off       | 0.00             | 11.0                         | OK            |
| max-Q gust     | 0.10 | <b>ON</b> | 1.00             | 0.0                          | <b>RETIRE</b> |
| max-Q gust     | 0.10 | off       | 0.50             | 1.0                          | <b>RETIRE</b> |

- fatigue *accumulates across flights* —  $P(\text{survive } 20)$  now decays even at constant sub-critical load (no i.i.d. shortcut)
- thresholds from expected cost (vehicle:repair = 100:1  $\Rightarrow \alpha=0.02, \beta=0.48$ )





Stage 0 anomaly map  $\cdot$  Stage 2 posterior  $\cdot P(\text{survive } n)$  with fatigue  $\cdot$  decision — single held-out CFRP field.

- **Stage 4 — fleet learning:** hierarchical Bayes over tail numbers; formalises the SpaceX fleet-leader practice (lead vehicle = likelihood, fleet = prior update)
- **Stage 5 — design feedback:** fleet damage statistics → layup/reinforcement optimisation (BO)
- **Sim-to-real:** flow-matching bridge ready for first measured full-field data (per-sample normalisation needs no healthy reference)
- **Targets:** JAXA reusable-vehicle programmes; journal paper (anomaly benchmark + calibrated FMPE + fatigue prognosis)

## One sentence

SpaceX learned this loop empirically over a decade — we formalise it so it works from the first inspection of the first article.